

1. Introduction

The Virtual Painter project leverages real-time hand gesture recognition to simulate painting on a digital canvas. It uses computer vision, machine learning, and OpenCV's MediaPipe library to detect and interpret hand movements. This technical report outlines the project's architecture, algorithms, constants, limitations, and recommendations.

2. System Architecture

The system architecture consists of several key components:

- Camera Input: Captures live video feed.
- Hand Tracking Module: Uses MediaPipe for real-time landmark detection.
- Gesture Recognition: Identifies specific gestures such as drawing, erasing, undo/redo, and color selection.
- Canvas Processing: Manages drawing operations, line smoothing, and adaptive erasing.
- UI Layer: Displays the virtual canvas with interactive buttons and gesture-based control.
- Performance Optimization: Employs GPU acceleration for smoother frame rates.

3. Key Constants and Parameters

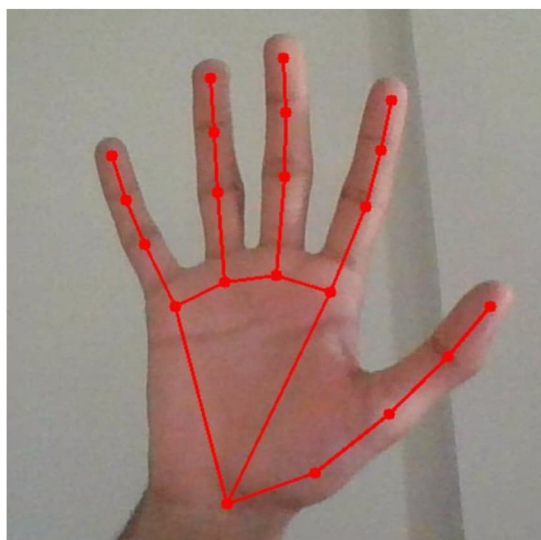
Some of the core parameters used in the system include:

- Brush Thickness: Dynamically adjusted based on finger distance.
- Eraser Size: Adaptive, depending on the palm openness.
- Frame Rate: Targeted at 60 FPS for minimal latency.
- Gesture Confidence Threshold: Used to ensure stable recognition before triggering actions.

4. Core Algorithms

The Virtual Painter uses the following algorithms:

1. Hand Detection Algorithm — Detects 21 key hand landmarks using MediaPipe.

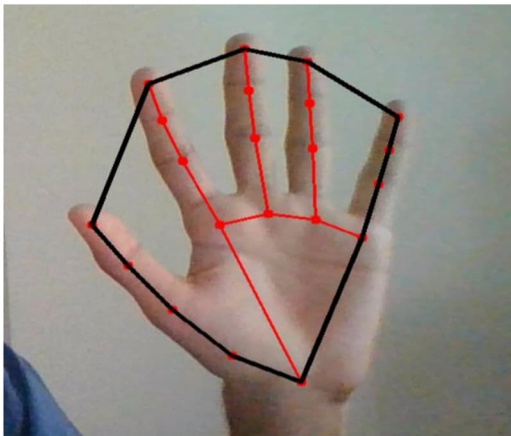


2. Gesture Mapping — Matches landmark configurations to specific gestures (draw, erase, color select).



3. Line Smoothing — Uses interpolation between successive finger coordinates for natural strokes.

4. Adaptive Erasing — Dynamically scales eraser size based on palm area and openness.



5. Undo/Redo System — Maintains an action stack and buffer for efficient canvas updates.

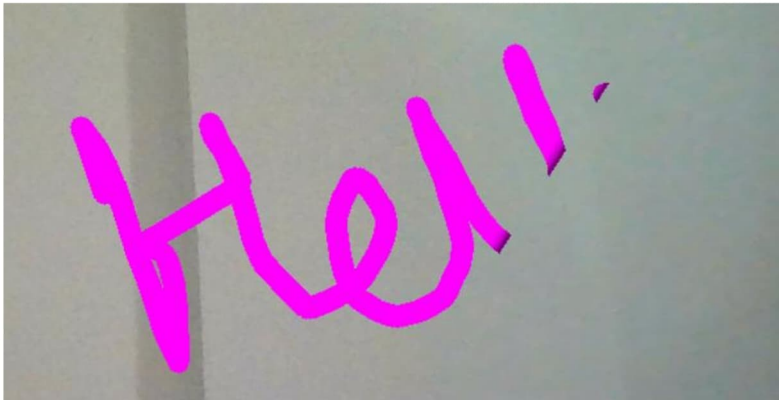
The program continuously monitors for keyboard input during execution to provide additional functionality and user control. The following key bindings are implemented:

- **'Q' Key:**
Terminates the main application loop and gracefully exits the Virtual Painter program.
- **'U' Key:**
Triggers the `undo()` function, allowing the user to revert the most recent drawing action on the canvas.
- **'R' Key:**
Invokes the `redo()` function to restore the previously undone action, thereby reapplying it to the canvas.
- **'S' Key:**
Saves the current state of the digital canvas as an image file.

The filename is automatically generated in the format Canvas_<timestamp>.png, ensuring each saved version is uniquely identified.

A confirmation message is printed to the console to inform the user of the successful save operation.

6. Edge Feathering — Adds soft borders during erasing for smoother results.



7. Canvas Saving— Integrated with backend for saving/exporting artworks and gesture data.



5. Limitations

Despite its performance, the Virtual Painter faces some limitations:

- High dependence on lighting and background uniformity.
- Occasional false gesture detections due to rapid hand motion.
- Limited accuracy for left-hand detection when overlapping with the right hand.
- Requires good camera quality for precise tracking.
- No built-in backend for saving sessions or user profiles.

6. Testing and Validation

The system underwent several test phases:

1. Functional Testing — Verified drawing, erasing, color selection, and undo/redo.
2. Performance Testing — Checked for lag and FPS stability under different lighting conditions.
3. Gesture Accuracy Testing — Compared predicted gestures vs. intended ones.
4. Compatibility Testing — Evaluated performance across different cameras and systems.

7. Recommendations and Future Improvements

To enhance usability and expand the Virtual Painter's features, the following improvements are recommended:

- Implement AI-based gesture prediction for smoother transitions.
- Introduce multi-user collaborative drawing.
- Add voice-assisted color selection.
- Enhance GPU acceleration for real-time rendering and higher-resolution outputs.

8. Conclusion

The Virtual Painter demonstrates a successful integration of computer vision and interactive design. With continued optimization and the inclusion of new AI-powered features, it holds potential for educational, artistic, and creative applications, showcasing the power of gesture-based interaction.